



THE UNIVERSITY OF
MELBOURNE

Marketing Research

Week 9





Housekeeping: About the Exam

- Exam weighting: the exam accounts for 50% of your final grade
- Duration of the exam: The exam will have a duration of 2 hours, and an additional 15 minutes allocated for reading time.
- Exam type: On-campus digital exam
- Exam format: LMS Quiz with lockdown browser. Ensure that you have downloaded the browser on your computer for the exam.
- Your task: The exam will consist of MCQs and long-answer type questions. You need to answer all questions.
- Authorised materials: You are permitted to bring the textbook, paper-based notes, and a non-programmable calculator without added text ability to the exam venue.



Housekeeping: Looking forward

- Week 9: Association + Correlation
- Week 10: Regression
- Week 11: Data visualization + (Principal Component Analysis or Cluster Analysis) + Group work day
- Week 12: Review



Recap

1.	Formulate the hypothesis
2.	Choose the significance level
3.	Select an appropriate test and check assumptions
4.	Calculate the test statistic
5.	Make the test decision
6.	Interpret the results

Type of Tests	Purpose of Test
One Sample T-Test	Compare the mean of a question answered by one group of subjects against a constant on metric data
Paired Samples T-Test	Compare two means of two different questions answered by the same group of subjects on metric data
Independent Samples T-Test	Compare the mean of a question answered by two different groups of subjects on metric data
ANOVA	Compare the means of a question answered by more than two different groups on metric data



Recap: Tests for Means

Compare a single mean to a test value (pre-specified by researcher)

- $H_0: \mu = \text{test value}$ $\leftarrow$ univariate
- One sample t-test

Compare two means

- $H_0: \mu_1 = \mu_2$ $\leftarrow$ bivariate
- If μ_1 and μ_2 are from the same population: paired samples t-test
- If μ_1 and μ_2 are from two different populations: independent samples t-test

Compare more than two population means (m groups)

- $H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_m$ $\leftarrow$ multivariate
- One-way ANOVA



Topics for today

1. Testing for Associations

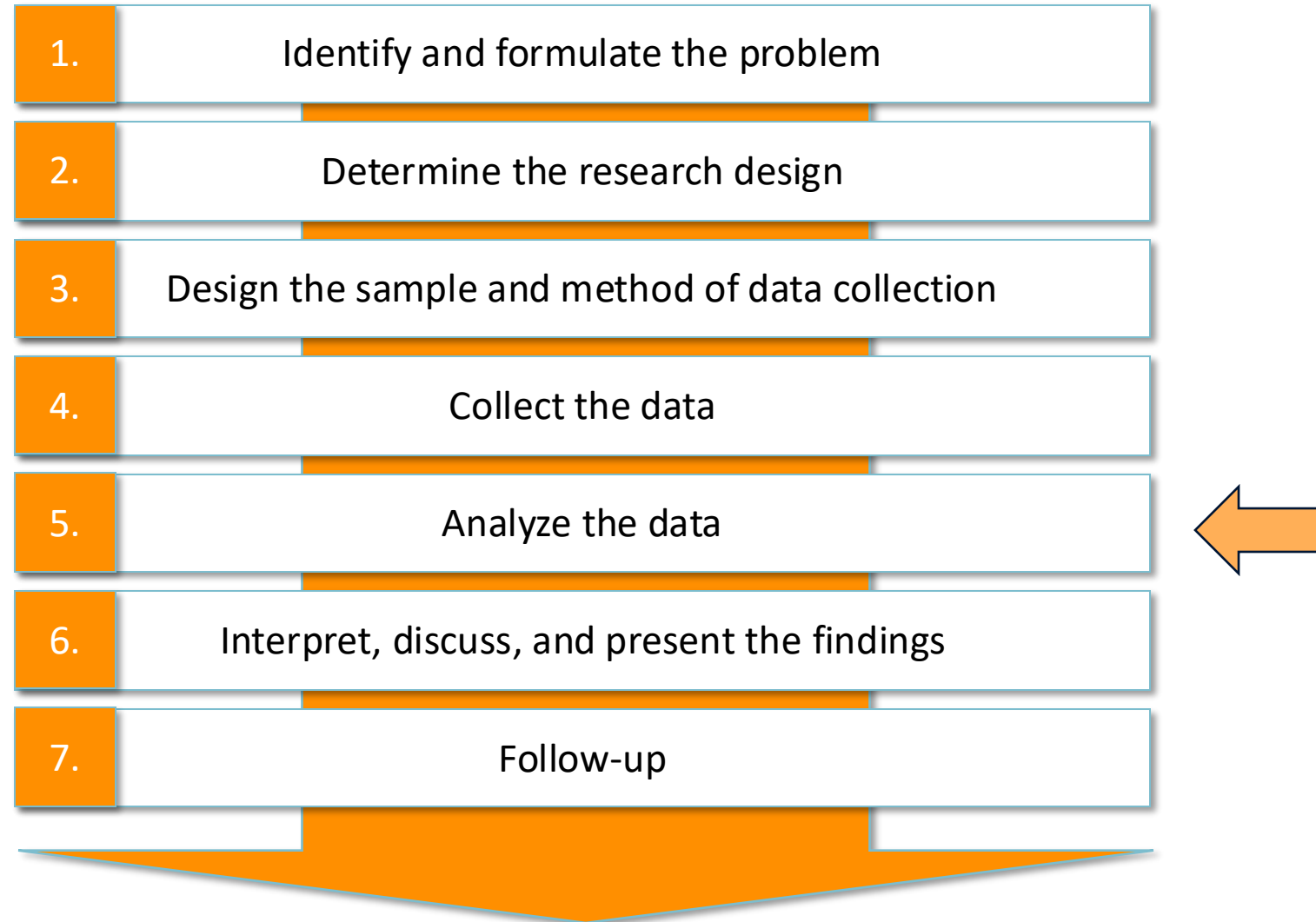
- Pearson product moment correlation coefficient
- Spearman rank order correlation coefficient
- Chi-square test of association

2. Activities

3. Bivariate Regression (if time permits)

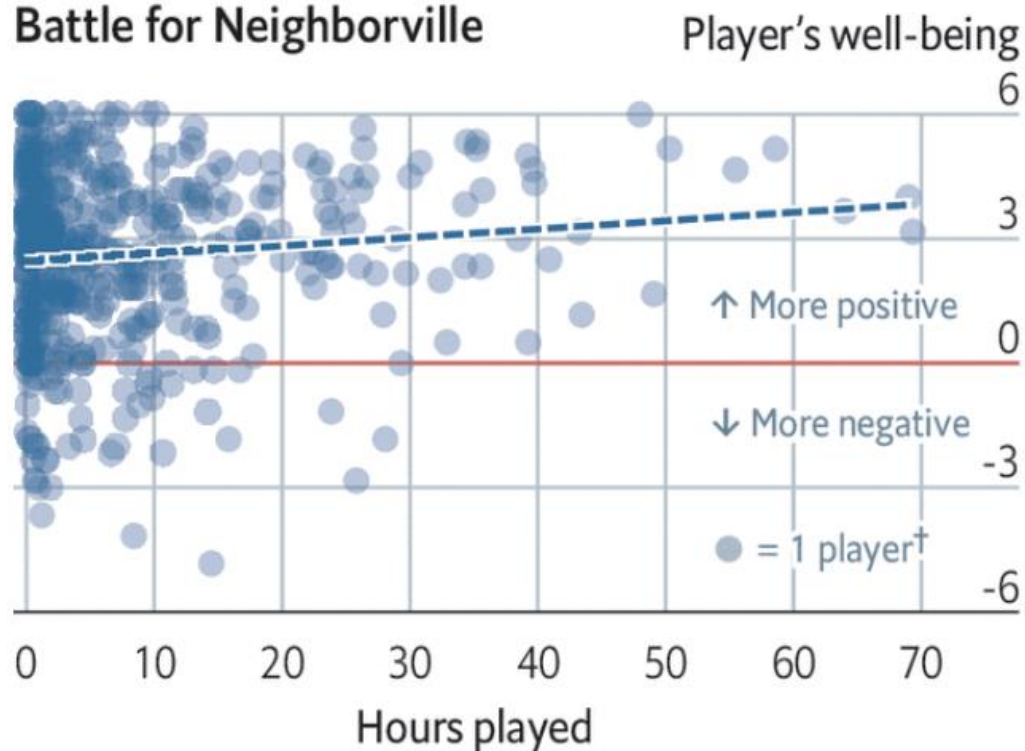


Testing for Associations



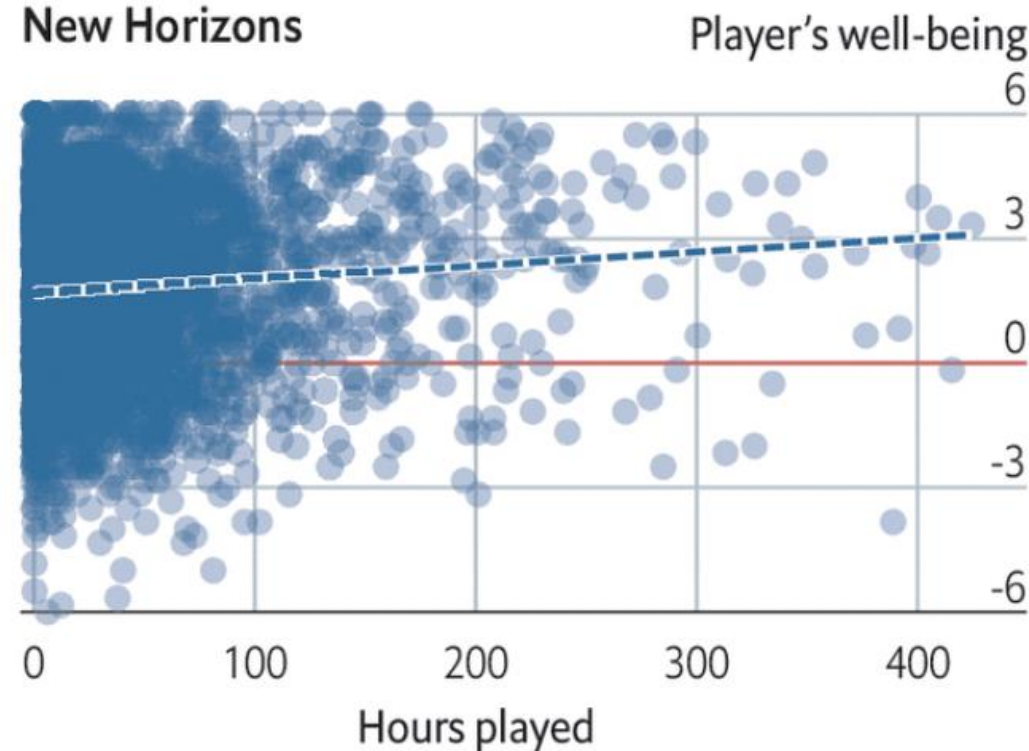
Playing video games in lockdown can be good for mental health

Plants vs. Zombies: Battle for Neighborville



Source: "Video game play is positively correlated with well-being", by N. Johannes, M. Vuorre and A. Przybylski, PsyArXiv 2020

Animal Crossing: New Horizons



*Net well-being score after two weeks. 6=most positive, -6=most negative

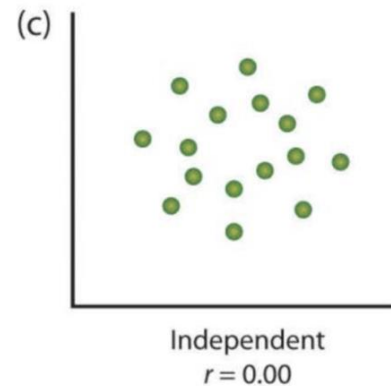
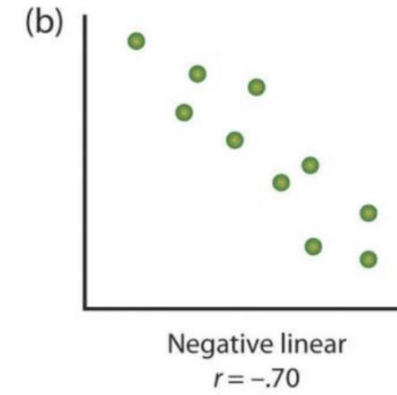
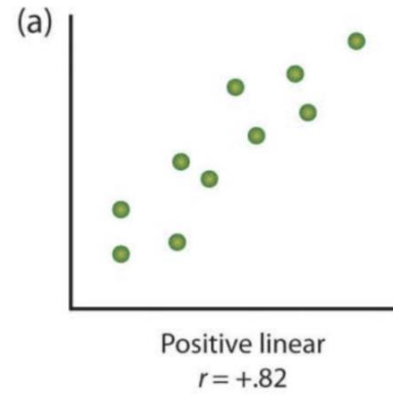
†Adult players in the United States, Canada and Britain



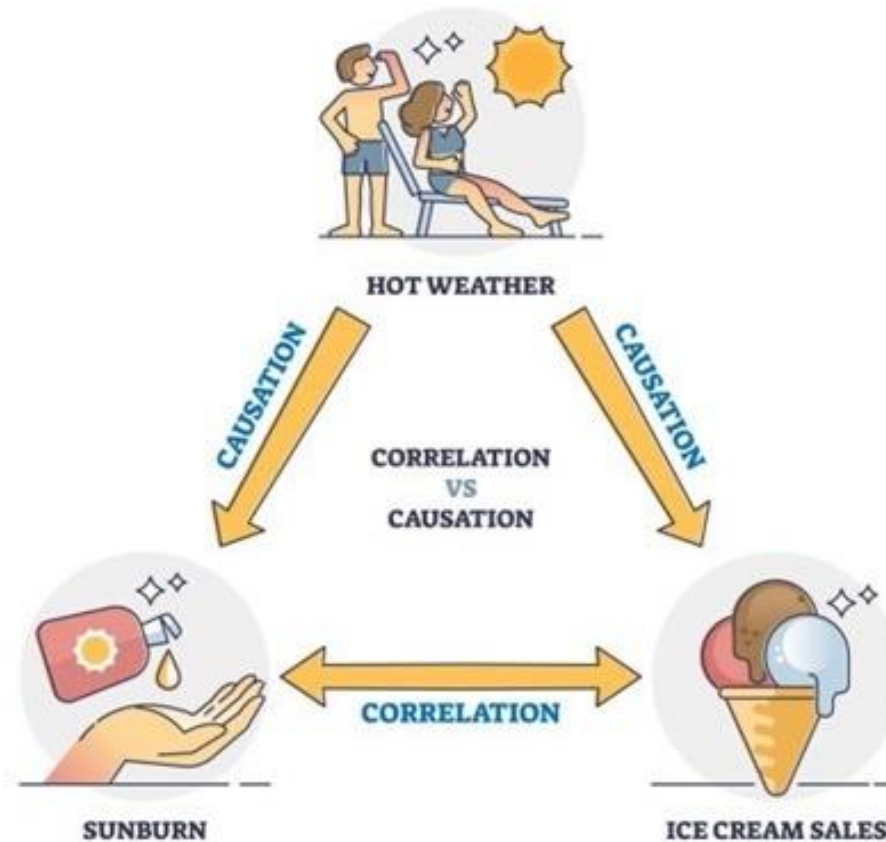
Testing for Associations

Features of describing an association

- Presence of association
- Direction of association
- Strength of association



Correlation $\neq$ Causation





Testing for Associations (Part I)

Types of Test	Purpose of Test
Chi-Square Analysis (χ^2)	Test for relationship between two <u>nominal</u> variables.
Spearman Rank Order Correlation Coefficient	Test for relationship between two variables that are not nominal and that at least one is <u>ordinal</u> .
Pearson Product Moment Correlation Coefficient	Test for relationships between two <u>interval and/or ratio</u> variables.



Testing for Associations (Part I)

	Nominal	Ordinal	Interval/Ratio
Nominal	Chi-square		
Ordinal	Chi-square	Spearman	
Interval/Ratio		Spearman	Pearson



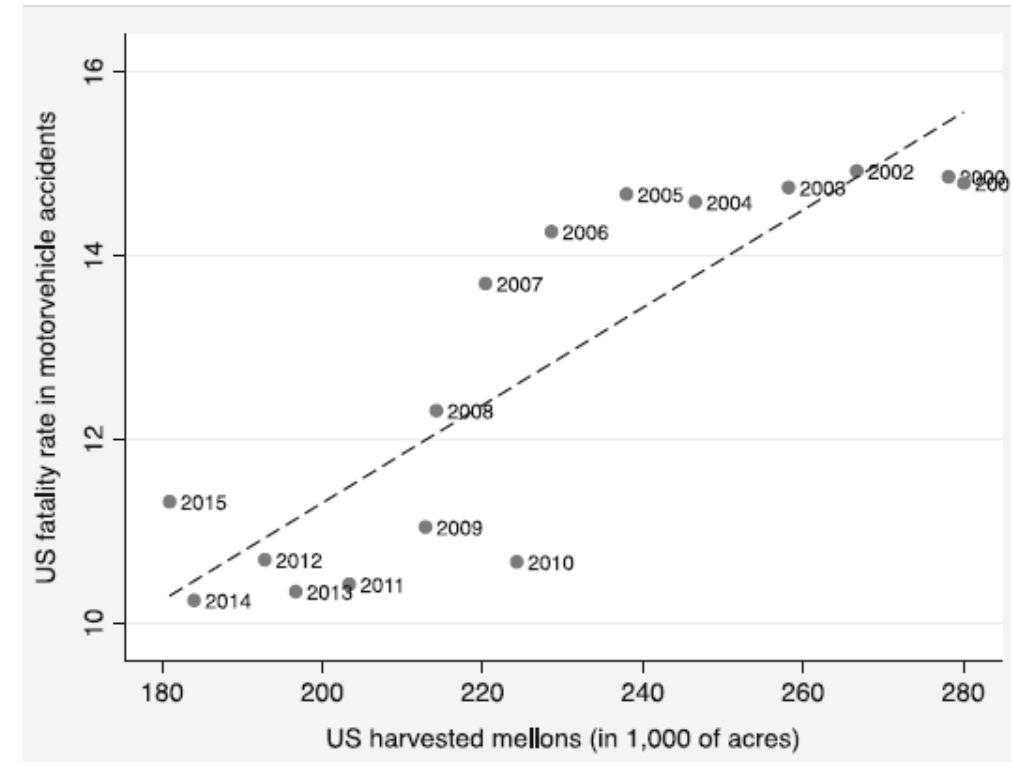
Correlation

❑ Explore the strength and direction of the **linear relationship** between two variables.

- Correlation indicates association NOT causation.
- correlation coefficient (r) shows the strength of association between two metric variables
 - below 0.30 indicates a weak relationship
 - between 0.30 and 0.49 indicates a moderate relationship
 - above 0.49 indicates a strong relationship

Example

- How strongly are sales related to advertising expenditures?
- Is there an association between market share and the size of the sales force?





Pearson Product Moment Correlation

Assumption

- Level of measurement (metric: Interval/ratio)
- All variable information must be from the same subject
- Independence of observations
- Linear relationship between variables



Pearson Product Moment Correlation

- Generate scatterplots to inspect for violations and nature of relationship (using SPSS)
 - **Graphs -> Scatter/Dot**
 - Dependent variable (DV) along the Y axis
 - Independent variable (IV) along the X axis
- Check the distribution of the data points
 - Is the cluster shape even (homoscedasticity)
 - Can you draw a straight line through the main cluster of points
 - Direction pointed by the line (upward or downward)

Pearson Product Moment Correlation

11	nps	Numeric	8	0	How likely is it that you would recommend our company to a friend or colleague?	{1, very unli...}
----	-----	---------	---	---	---	-------------------

13	sat1	Numeric	8	0	In general, I am satisfied with Oddjob Airways.	{1, Fully dis...}
----	------	---------	---	---	---	-------------------

Value ▾	Label
1	very unlikely
2	1
3	2
4	3
5	4
6	5
7	6
8	7
9	8
10	9
11	very likely

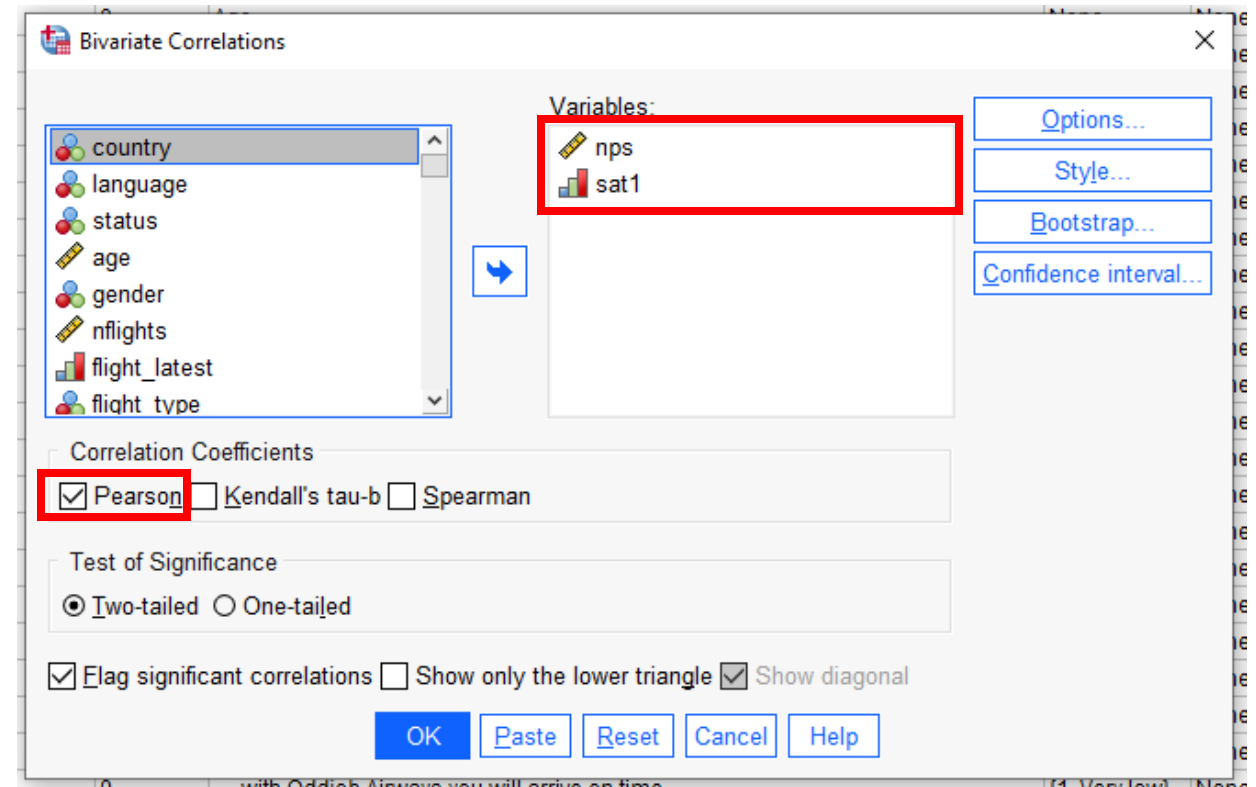
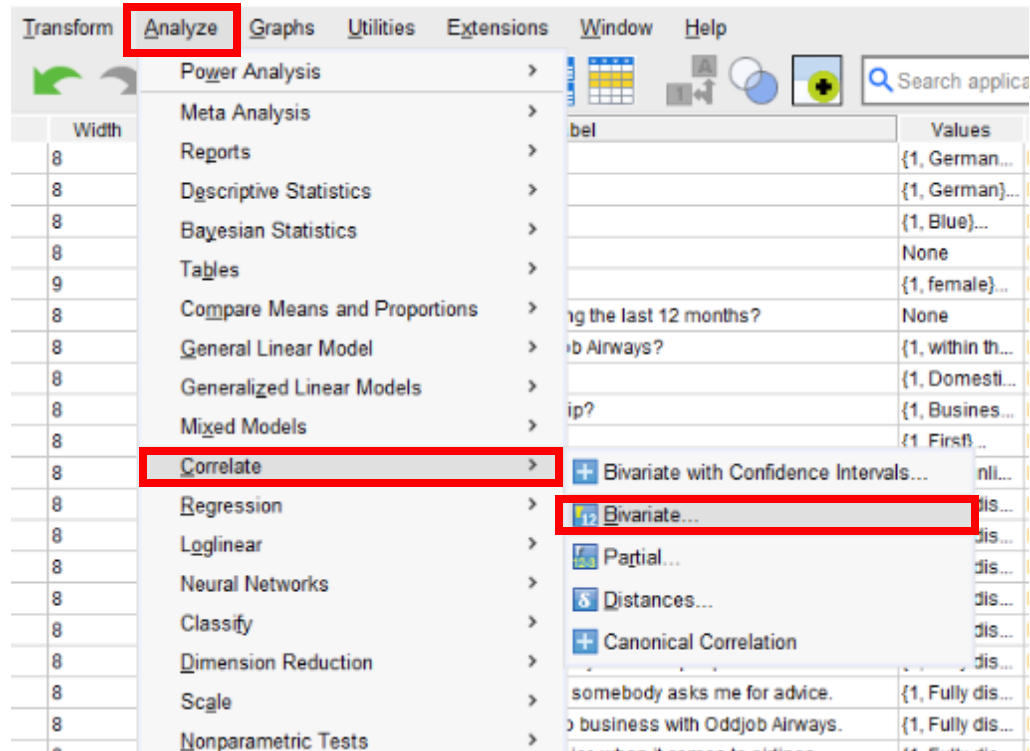
Value ▾	Label
1	Fully disagree
7	Fully agree

- Two interval scale variables → Pearson Product Moment Correlation
- Significance level: 5%
- Testing variables: Willingness to recommend (nps), General satisfaction (sat1)
- Hypothesis: There is an association(correlation) between willingness to recommend and customer satisfaction with Oddjob Airways.



Pearson Product Moment Correlation

Analyze → Correlate → Bivariate → Select variables of interest → Tick 'Pearson' and 'Two-tailed'



Pearson Product Moment Correlation

Correlations

		How likely is it that you would recommend our company to a friend or colleague?	In general, I am satisfied with Oddjob Airways.
How likely is it that you would recommend our company to a friend or colleague?	Pearson Correlation	1	.771**
	Sig. (2-tailed)		<.001
	N	1065	1065
In general, I am satisfied with Oddjob Airways.	Pearson Correlation	.771**	1
	Sig. (2-tailed)	<.001	
	N	1065	1065

Pearson Correlation Coefficient (r)

** . Correlation is significant at the 0.01 level (2-tailed).

H_0 : There is NO association/correlation between NPS and overall satisfaction for Oddjob Airways.

H_1 : There is an association/correlation between NPS and overall satisfaction for Oddjob Airways.

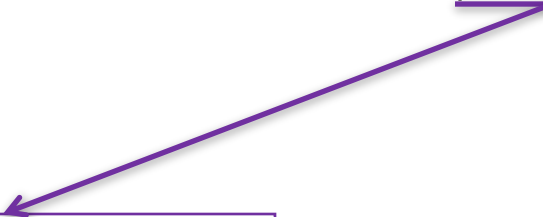


Pearson Product Moment Correlation

How to report the results of Pearson product moment correlation?

Example

In this test, we hypothesized that there is an association between willingness to recommend and customer satisfaction with Oddjob Airways. The result of a Pearson Product Moment Correlation test shows a significant and positive correlation between willingness to recommend and satisfaction ($r = .771$, $p < .001$). Thus, the hypothesis is supported.

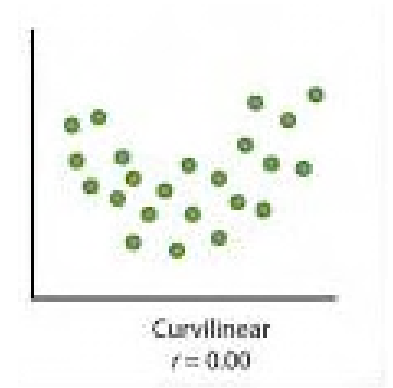


r = correlation coefficient

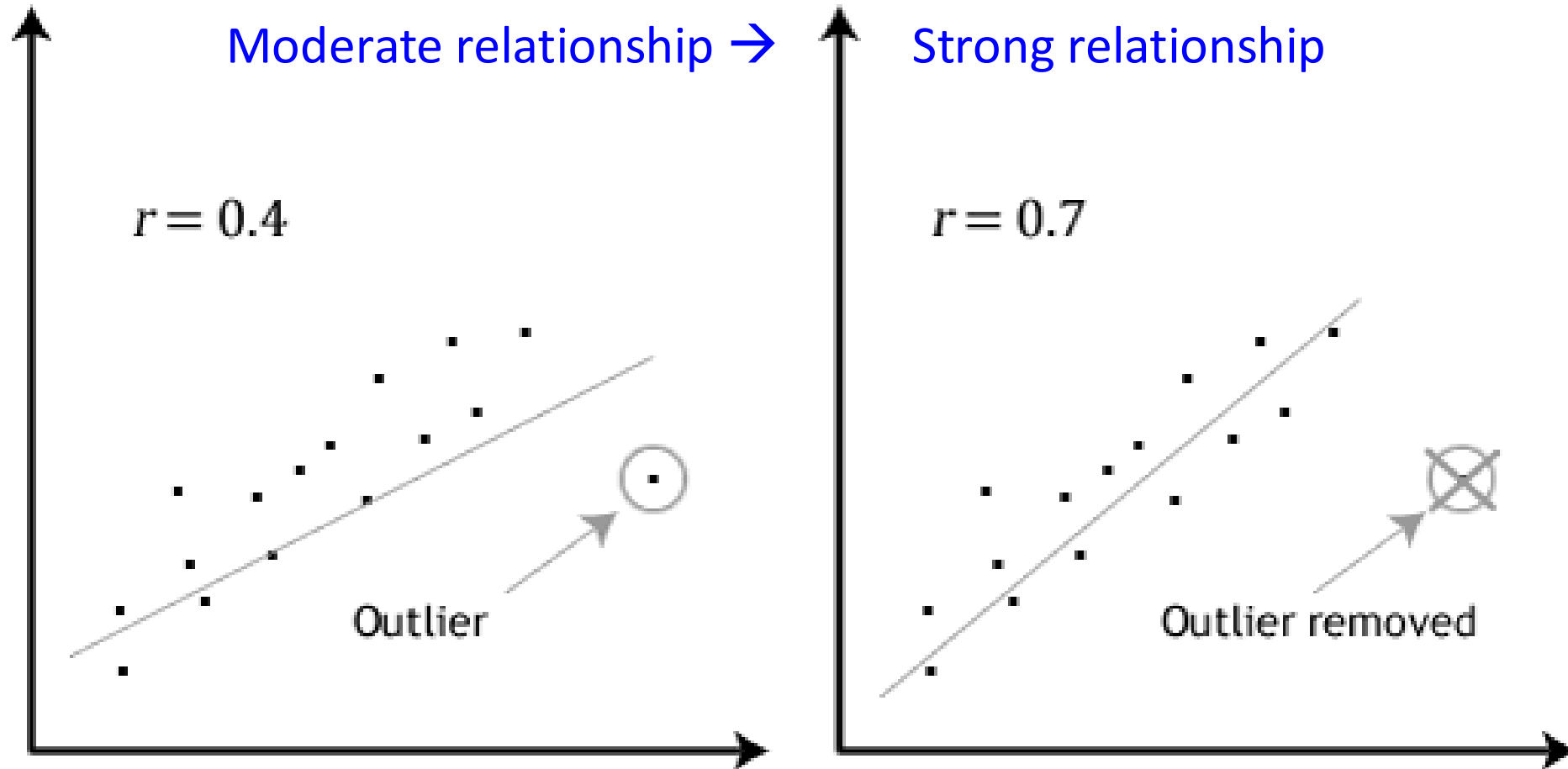


Important Notes about (Pearson) Correlation Coefficient: r

- r describes a "linear relationship" between two variables
 - $r = 0$: **NO** linear relationship
 - **However, there could still be a curvilinear relationship:**
 - $r = \pm 1$: a **straight line** can describe the relationship
- Coefficient of determination (r^2) = $r * r$
 - % of the variation in one variable can be accounted by the other one
 - E.g., $r^2 = 0.36$ ($r = 0.6$) means that 36% of the variation in one variable can be explained by the other variable
 - Similar idea as "R-square" in a bivariate regression model (with one IV; one DV)



Outlier's impact on Pearson Correlation



The Pearson correlation coefficient is easily affected by outliers. Why?



Important Notes about Correlation Coefficient: r

- Correlation does NOT indicate causality
 - E.g. Increase in Christmas card sales does not cause Christmas;
- Statistical significance is NOT the same as substantive significance (or economic significance, practical significance).
 - E.g. Unit sales = $100 + 0.0005 * \text{Ad Spending in \$}$
 - Every \$10,000 spending will lead to 5 more units sales
 - If the unit price is low, then little substantive significance
- Correlation is NOT interaction
 - Interaction: TWO independent variables act together to affect your dependent variable. E.g., the attitude (Y) toward the advertisements (X1) varies with the mood (X2) and the interaction ($X1 * X2$)



Spearman Rank Order Correlation

- ❑ Measure the strength and direction of a relationship between two variables that are *not nominal* and that *at least one is ordinal*.

Example

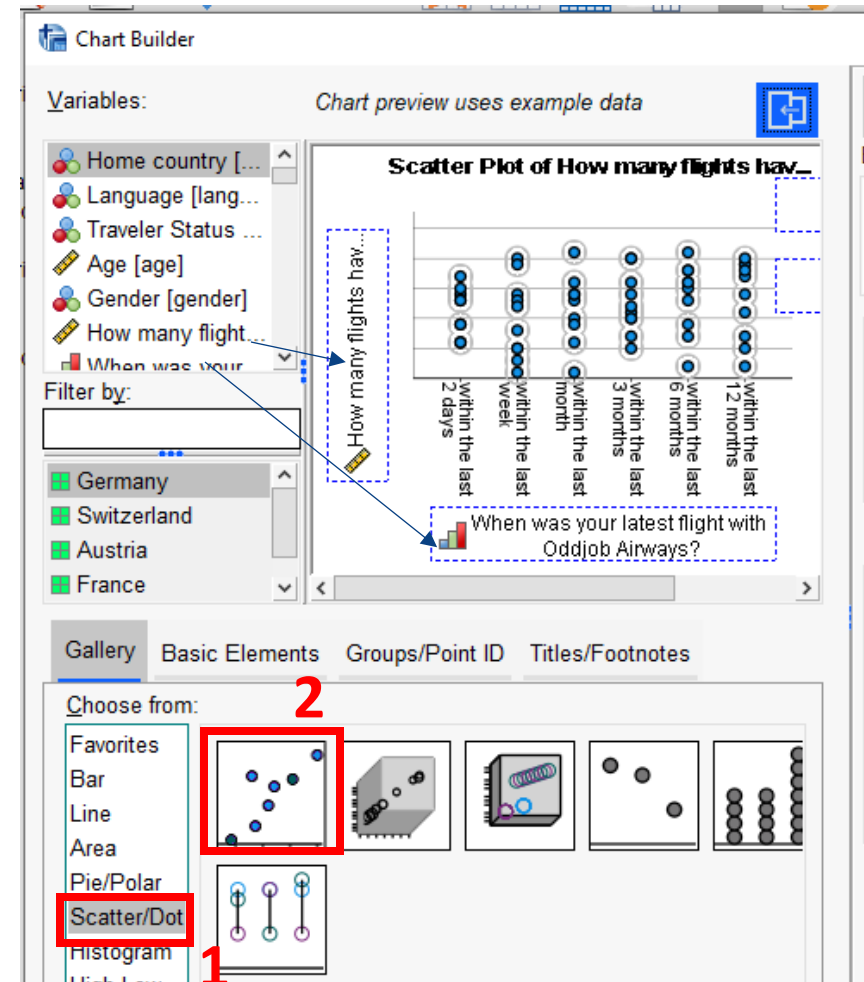
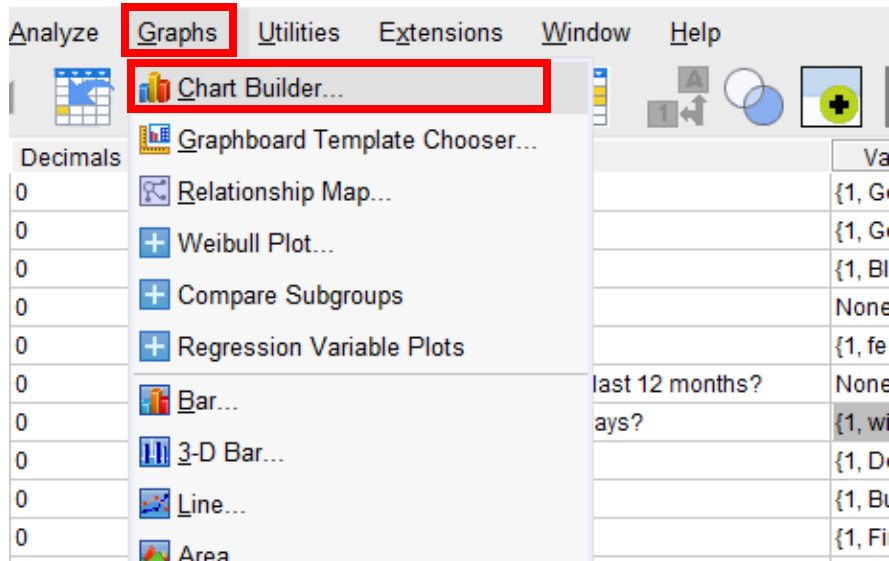
6	nflights	Numeric	8	0	How many flights have you taken during the last 12 months?	None
7	flight_latest	Numeric	8	0	When was your latest flight with Oddjob Airways?	{1, within t...

Value ▾	Label
1	within the last 2 days
2	within the last week
3	within the last month
4	within the last 3 months
5	within the last 6 months
6	within the last 12 months

- One ratio, one ordinal variables → Spearman rank order correlation
- Significance level: 5%
- Testing variables: number of flights taken (nflights), latest flight (flight_latest)
- Hypothesis: There is an association between the number of flights taken during the last 12 months and the most recent flight taken with Oddjob Airways.

Spearman Rank Order Correlation

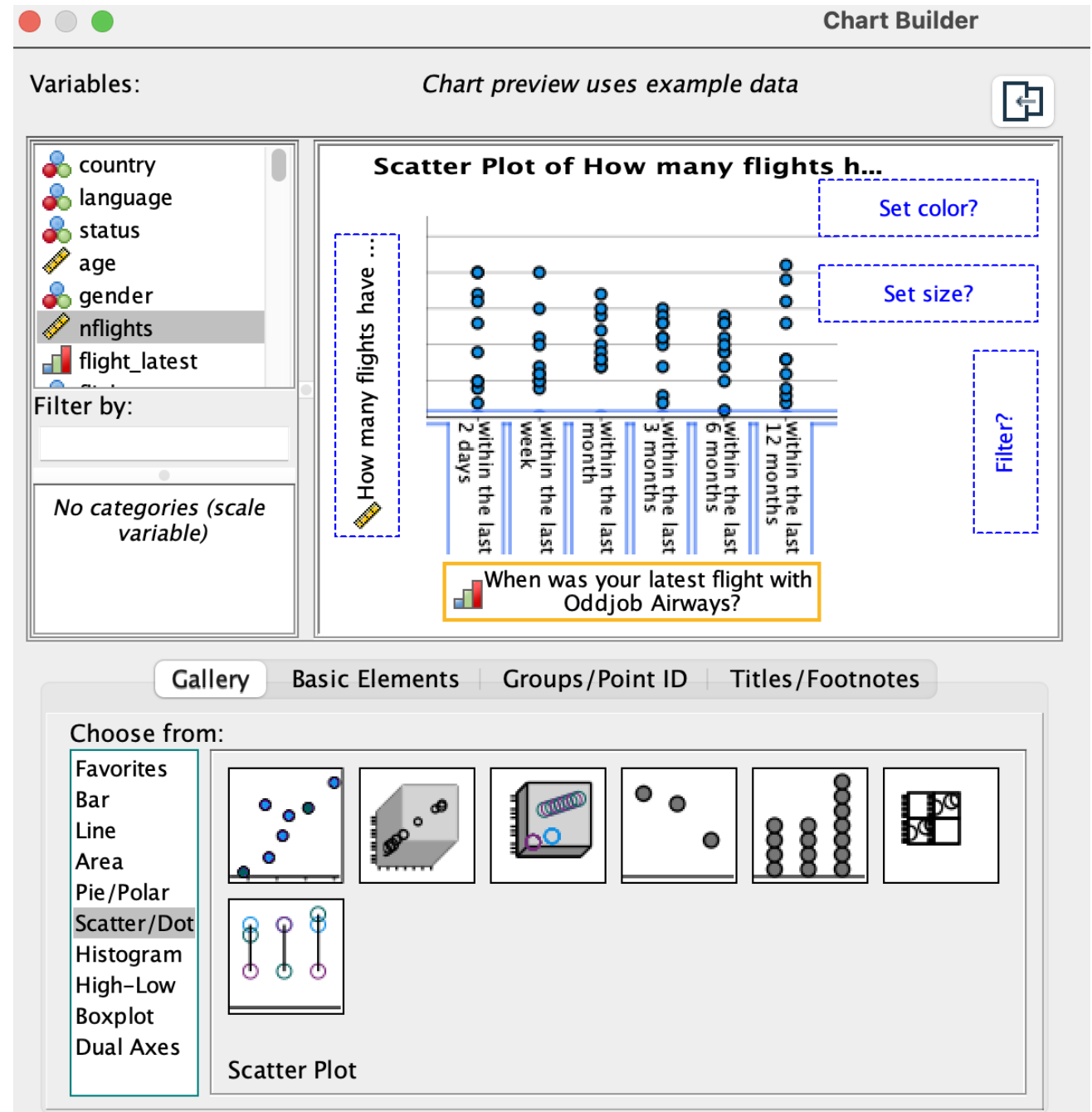
Preliminary Analysis: Graphs → Chart Builder → Scatter/Dot → Scatter Plot





Scatter plot

- Choose from “Scatter/Dot”
- Once the chart preview shows up:
 - *nflights* to Y axis
 - *flight_latest* to X axis



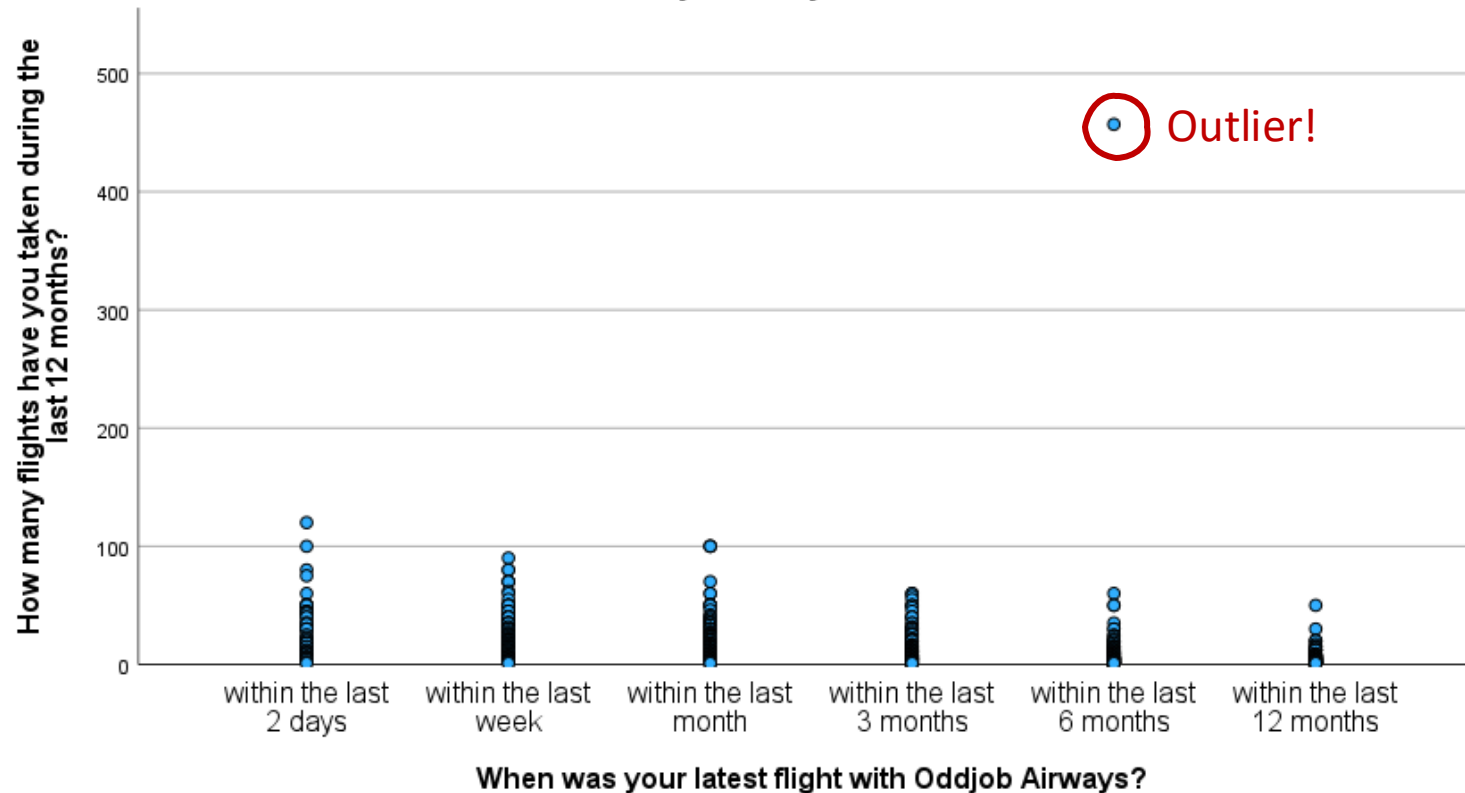


Spearman Rank Order Correlation

Tip – Create ID

Transform → Compute Variable → Enter 'ID' under Target Variable → double click '\$Casenum' under Functions and Special Variables

Scatter Plot of How many flights have you taken during the last 12 months? by When was your latest flight with Oddjob Airways?



Outlier: ID = 935, whose z-score for the number of flights is 21.93 (above 3). Considering that this individual took 457 flights, with the most recent one occurring within the last 6 months, it is evident that this outlier is likely due to a data entry error. Since the original data is missing, the data needs to be deleted.

Tip – Keep record in Syntax

ex) `*nflights outlier ID=935, data deleted*`



Dealing with the outlier(s) for nflights

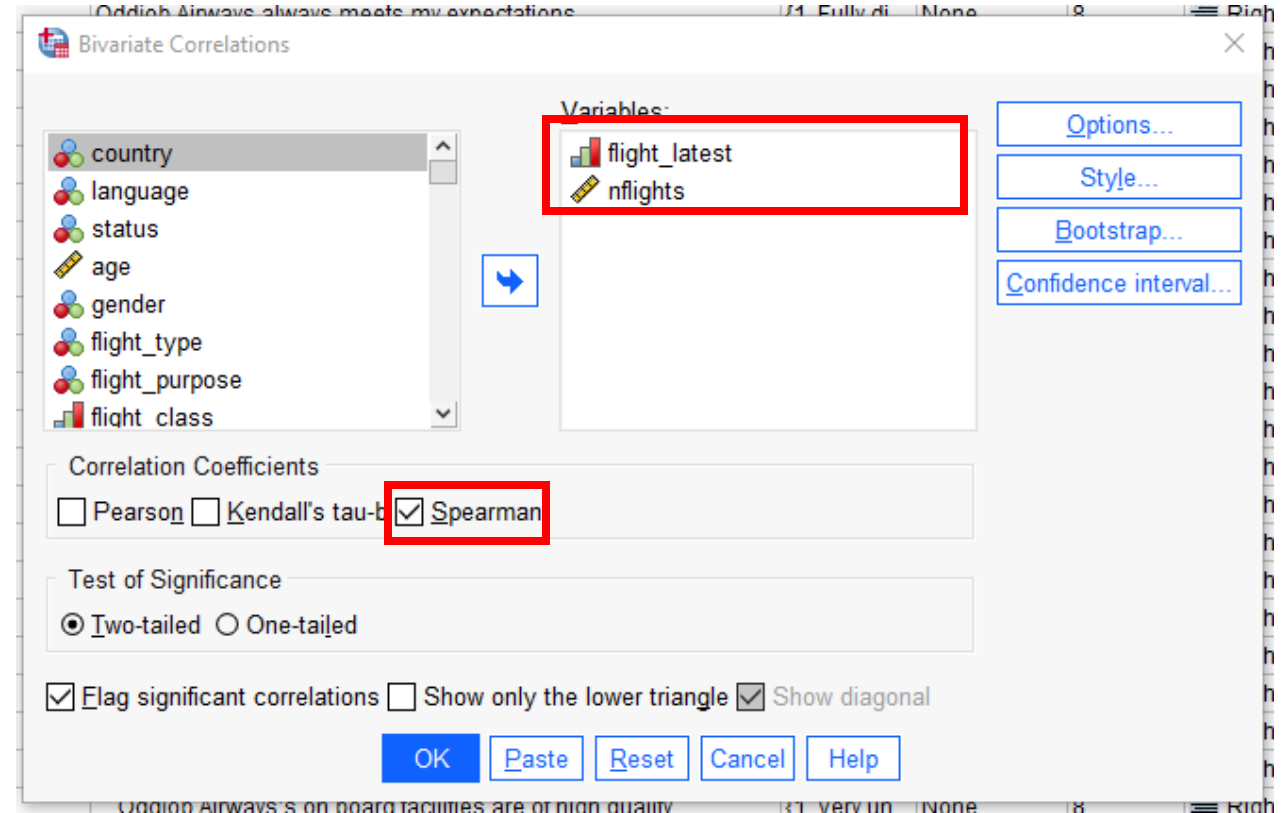
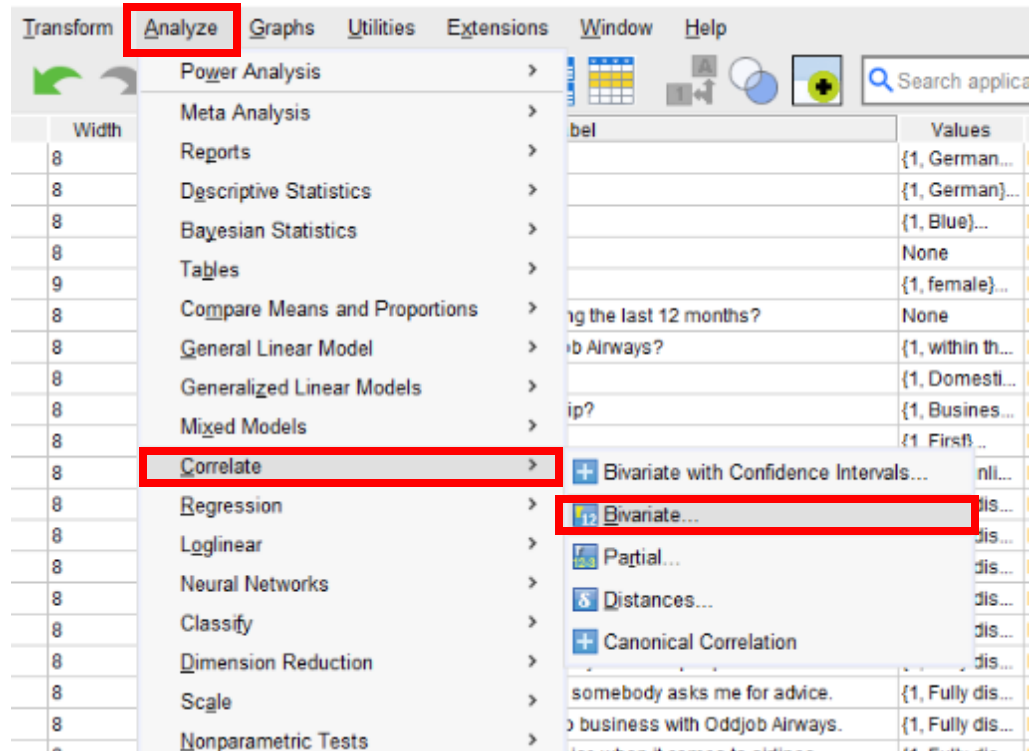
- Note: To properly check the outliers, use the Z value for nflight, “Znflight” (= demeaned variable /SD of nflight)
- *Analyze -> Descriptive statistics -> Descriptive -> Select ‘nflights’-> Check ‘save standardized values as variables’*
- *Analyze -> Descriptive Statistics -> Frequencies -> Check the relevant statistics in “Statistics”.*
- You can clear the outlier. However, make a copy of your original dataset before making any changes to it. (e.g., Transform → Compute Variable → Target Variable = ‘nflights_backup’ and Numeric Expression = ‘nflights’).

		How many flights have you taken during the last 12 months?	Zscore: How many flights have you taken during the last 12 months?
N	Valid	1065	1065
	Missing	0	0
Mean		13.42	.0000000
Median		8.00	-.2679053
Std. Deviation		20.226	1.0000000
Variance		409.110	1.000
Minimum		1	-.61399
Maximum		457	21.93073



Spearman Rank Order Correlation

Analyze → Correlate → Bivariate → Select variables of interest → Tick 'Spearman'



Spearman Rank Order Correlation

Correlations

			When was your latest flight with Oddjob Airways?	How many flights have you taken during the last 12 months?
Spearman's rho	When was your latest flight with Oddjob Airways?	Correlation Coefficient	1.000	-.497**
		Sig. (2-tailed)	.	<.001
		N	1065	1064
	How many flights have you taken during the last 12 months?	Correlation Coefficient	-.497**	1.000
		Sig. (2-tailed)	<.001	.
		N	1064	1064

** . Correlation is significant at the 0.01 level (2-tailed).

Note: this is done AFTER the outlier was removed.

H_0 : There is NO association/correlation between the number of flights during the last 12 months and the latest flight with Oddjob Airways.

H_1 : There is an association/correlation between the number of flights during the last 12 months and the latest flight with Oddjob Airways.



Spearman Rank Order Correlation

How to report the results of Spearman Rank Order Correlation?

Example

In this test, we hypothesized that there is an association between the number of flights taken during the last 12 months and the most recent flight taken with Oddjob Airways. The result of a Spearman Rank Order Correlation test shows a significant, negative correlation between these two variables ($r = -.497$, $p < .001$). Customers who have taken more flights in the last 12 months are more likely to have taken their most recent flight recently. Thus, the hypothesis is supported.



r = Spearman correlation coefficient



Chi-Square Analysis

- ❑ Examine whether *two nominal variables* are associated in the population
 - Summarize in a two-way frequency table, “cross-tabulation (crosstab)”
 - The test compares the observed frequencies of the responses with the expected frequencies
 - Does not assess direction of the relationship
 - Examples: the relationship between the following variables, including socio-demographic(e.g., sex, education groups, income groups), Ad exposure (Y/N), Purchase (Y/N), Brand choice (Coke/Pepsi/Dr. Pepper)

Assumptions for Chi-Square test

- Independence of observation
 - Each observation should be generated by a different subject and no subject is counted twice
- Size of expected frequencies
 - Ideally the minimum expected frequency for each variable/cell should be 5



Chi-Square Analysis

Example

1	country	Numeric	8	0	Home country	{1, German...}
3	status	Numeric	8	0	Traveler Status	{1, Blue}...

Value ▾	Label
1	Germany
2	Switzerland
3	Austria
4	France
5	USA

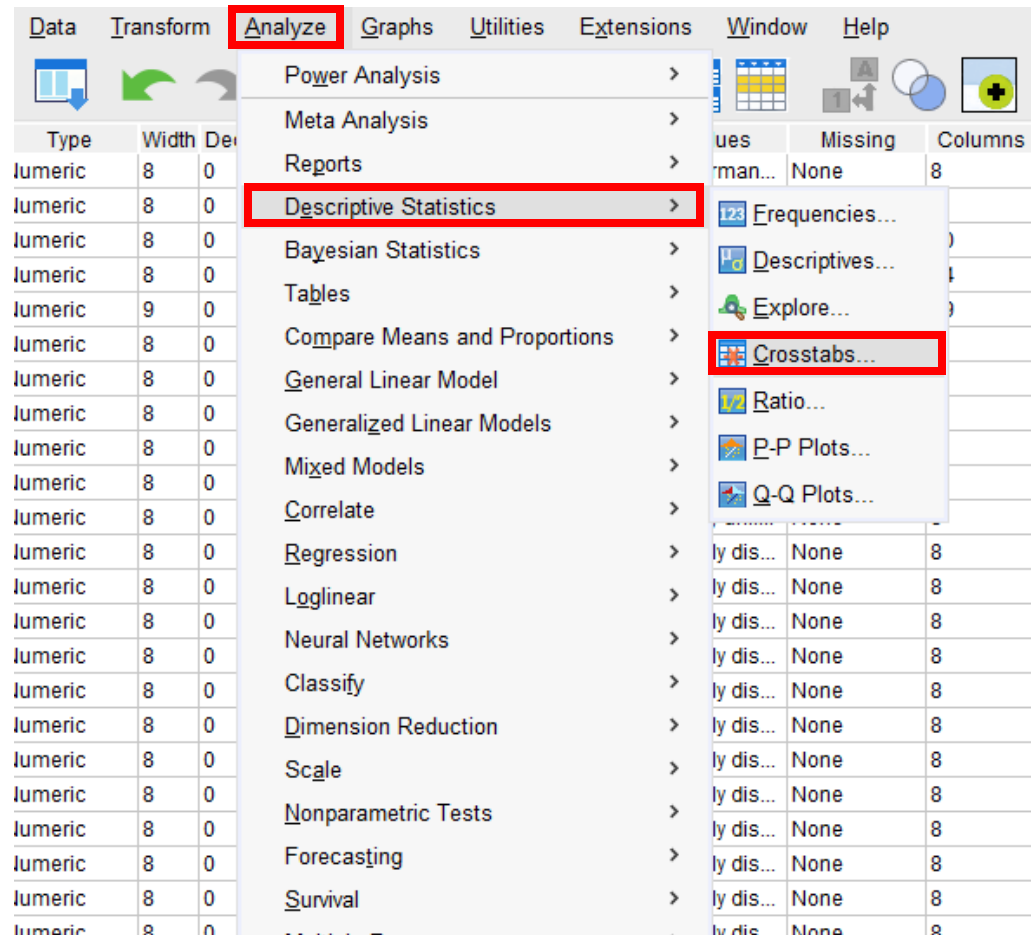
Value ▾	Label
1	Blue
2	Silver
3	Gold

- One nominal and One ordinal variables -> Chi-square analysis (it disregards the order though!)
- Significance level: 5%
- Variables: Home country (country), Traveler Status (status)
- Hypothesis: There is an association between the home country and the traveler's status.



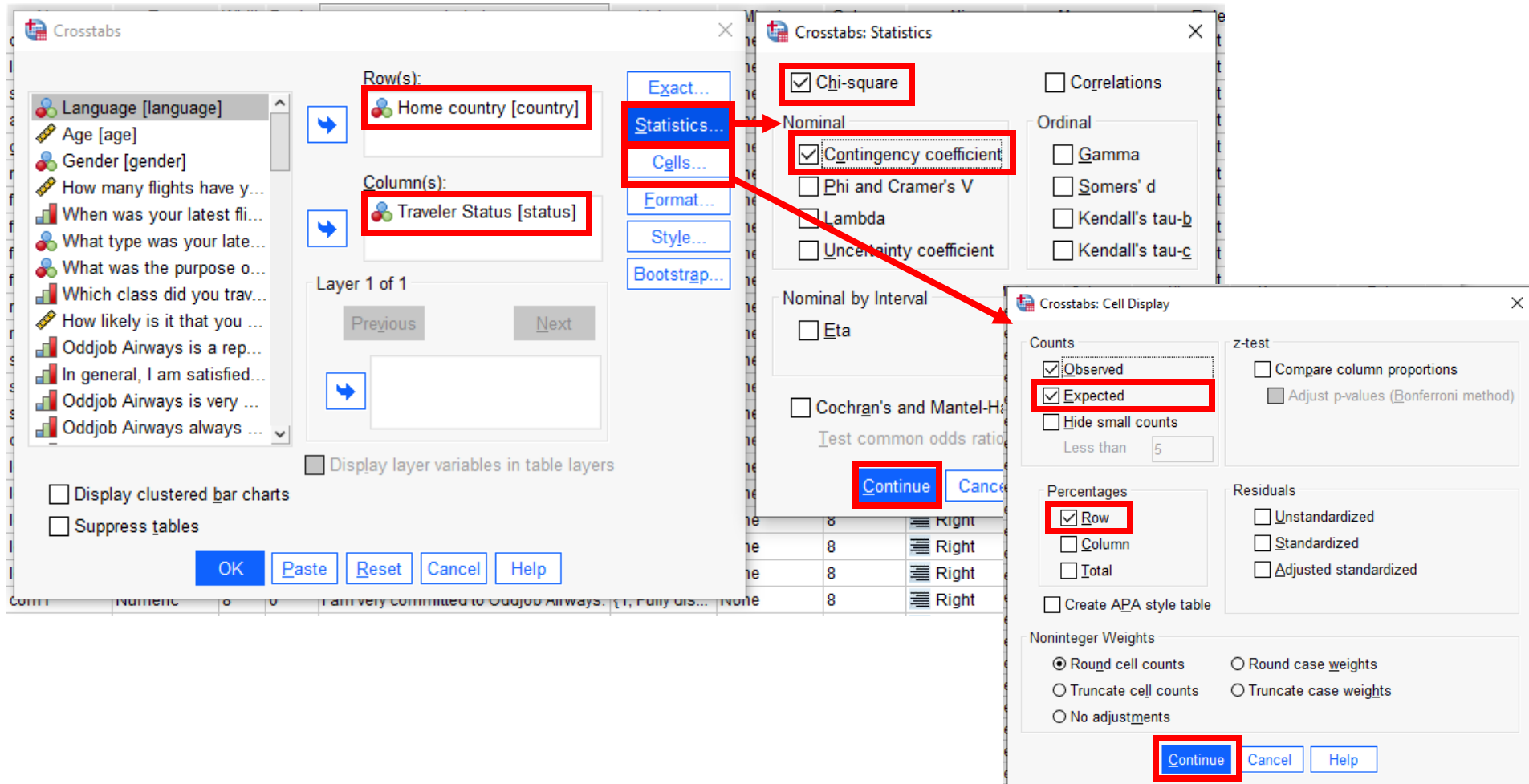
Chi-Square Analysis

Analyze → Descriptive Statistics → Crosstabs → One variable to Row → The other variable to Column(s) → [Statistics] → Tick 'Chi-square' and 'Contingency Coefficient' → continue → [Format] → Tick 'Expected' and 'Row' → Continue



Chi-Square Analysis

Analyze → Descriptive Statistics → Crosstabs → One variable to Row → The other variable to Column(s) → [Statistics] → Tick 'Chi-square' and 'Contingency Coefficient' → continue → [Cells] → Tick 'Expected' and 'Row' → Continue



The image displays three overlapping SPSS dialog boxes for a Chi-Square analysis, with red boxes and arrows highlighting the required steps:

- Crosstabs Dialog:**
 - Row(s):** Home country [country]
 - Column(s):** Traveler Status [status]
 - Buttons:** Statistics... (highlighted), Cells...
- Crosstabs: Statistics Sub-dialog:**
 - Nominal:** ☒ Chi-square, ☒ Contingency coefficient
 - Continue** button (highlighted)
- Crosstabs: Cell Display Sub-dialog:**
 - Counts:** ☒ Observed, ☒ Expected
 - Percentages:** ☒ Row
 - Continue** button (highlighted)

Chi-Square Analysis

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	36.577 ^a	8	<.001
Likelihood Ratio	39.031	8	<.001
Linear-by-Linear Association	4.008	1	.045
N of Valid Cases	1065		

a. 3 cells (20.0%) have expected count less than 5. The minimum expected count is .13.

Results *should not be interpreted* when more than 20% of the cells have an expected count of less than 5, or if any cell has an expected count of less than 1.

Chi-Square Analysis

Home country * Traveler Status Crosstabulation

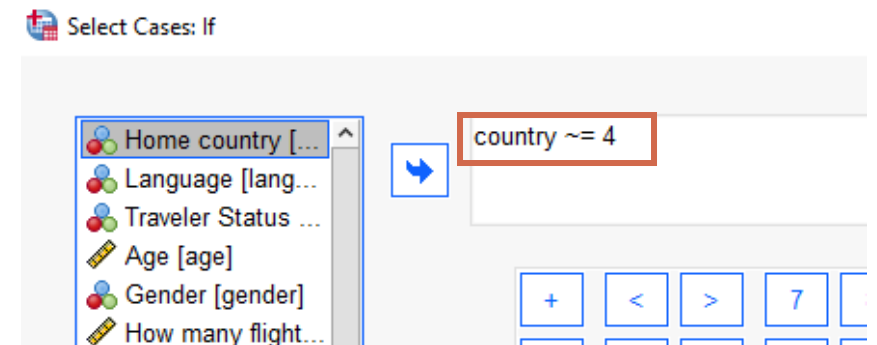
			Traveler Status			
			Blue	Silver	Gold	Total
Home country	Germany	Count	438	168	89	695
		Expected Count	441.8	159.9	93.3	695.0
		% within Home country	63.0%	24.2%	12.8%	100.0%
	Switzerland	Count	35	20	11	66
		Expected Count	42.0	15.2	8.9	66.0
		% within Home country	53.0%	30.3%	16.7%	100.0%
	Austria	Count	51	36	21	108
		Expected Count	68.7	24.8	14.5	108.0
		% within Home country	47.2%	33.3%	19.4%	100.0%
	France	Count	1	0	0	1
		Expected Count	.6	.2	.1	1.0
		% within Home country	100.0%	0.0%	0.0%	100.0%
	USA	Count	152	21	22	195
		Expected Count	124.0	44.9	26.2	195.0
		% within Home country	77.9%	10.8%	11.3%	100.0%
Total	Count	677	245	143	1065	
	Expected Count	677.0	245.0	143.0	1065.0	
	% within Home country	63.6%	23.0%	13.4%	100.0%	

There is only one French traveler, resulting in three cells with an expected count of fewer than 5.

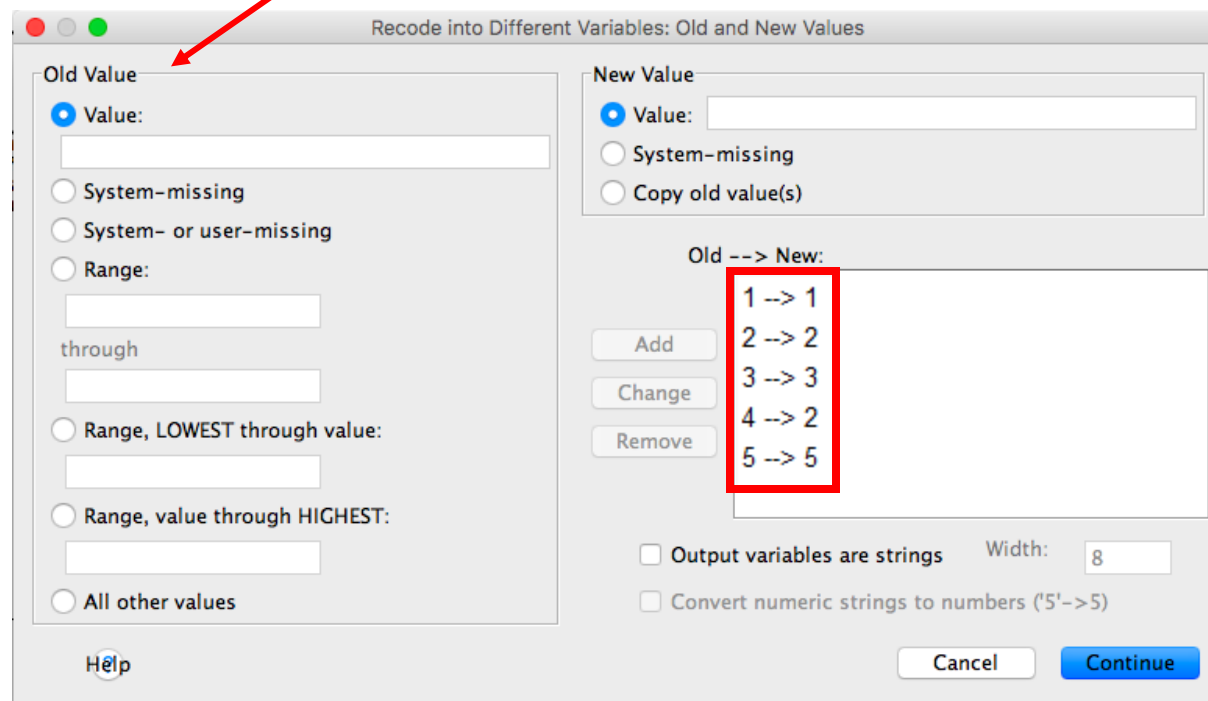
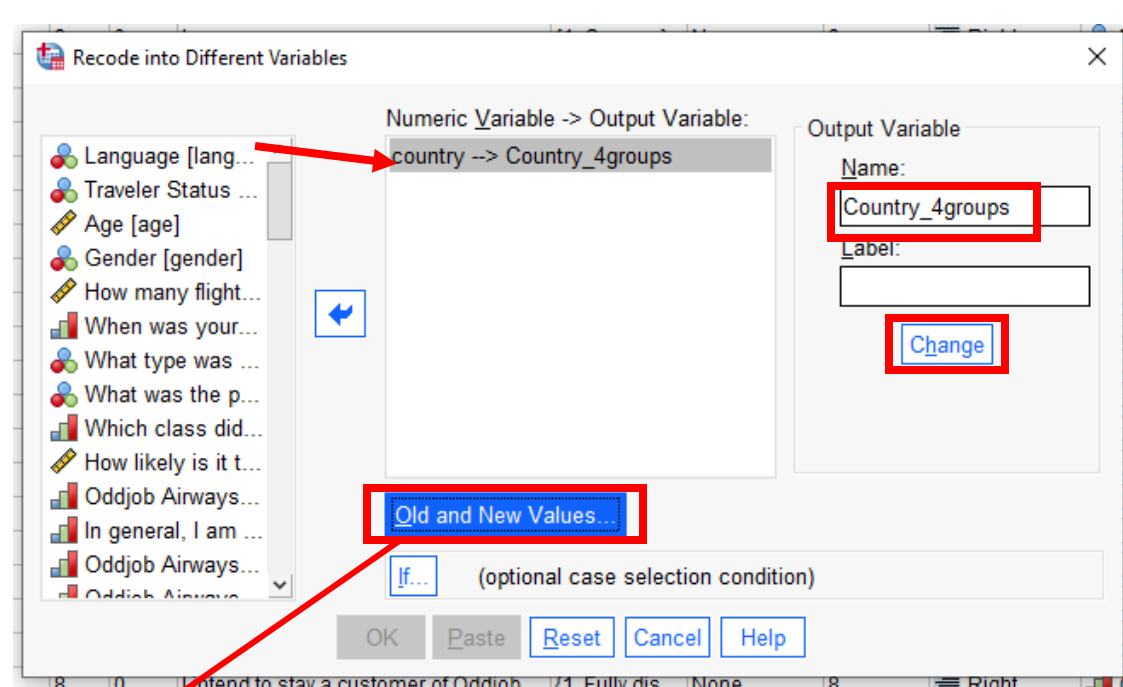
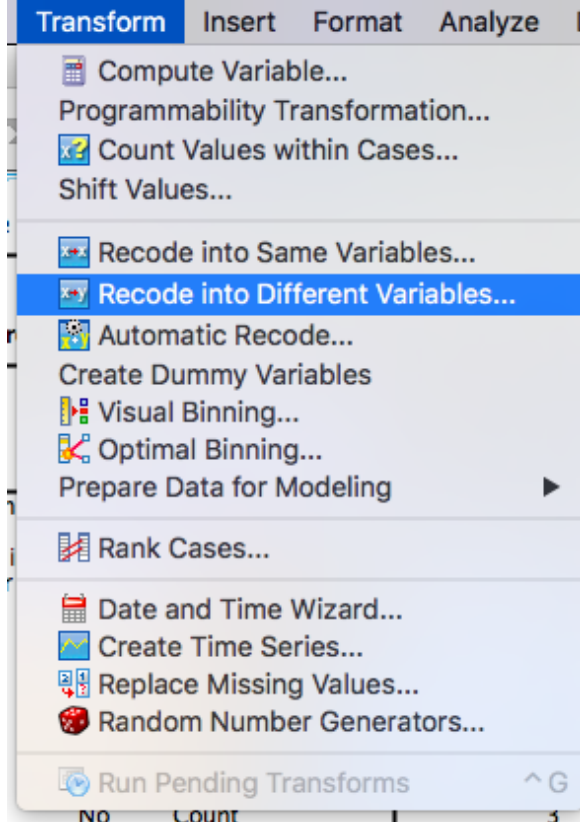
Solution 1: Exclude France from the analysis

(Data → Select Cases → Choose 'If condition is satisfied' → Click 'If' → double click 'Home country' → Enter “~= 4” → Continue)

*4 indicates France



Solution 2: Merge France with another country



Chi-Square Analysis

Value Labels

Spelling...

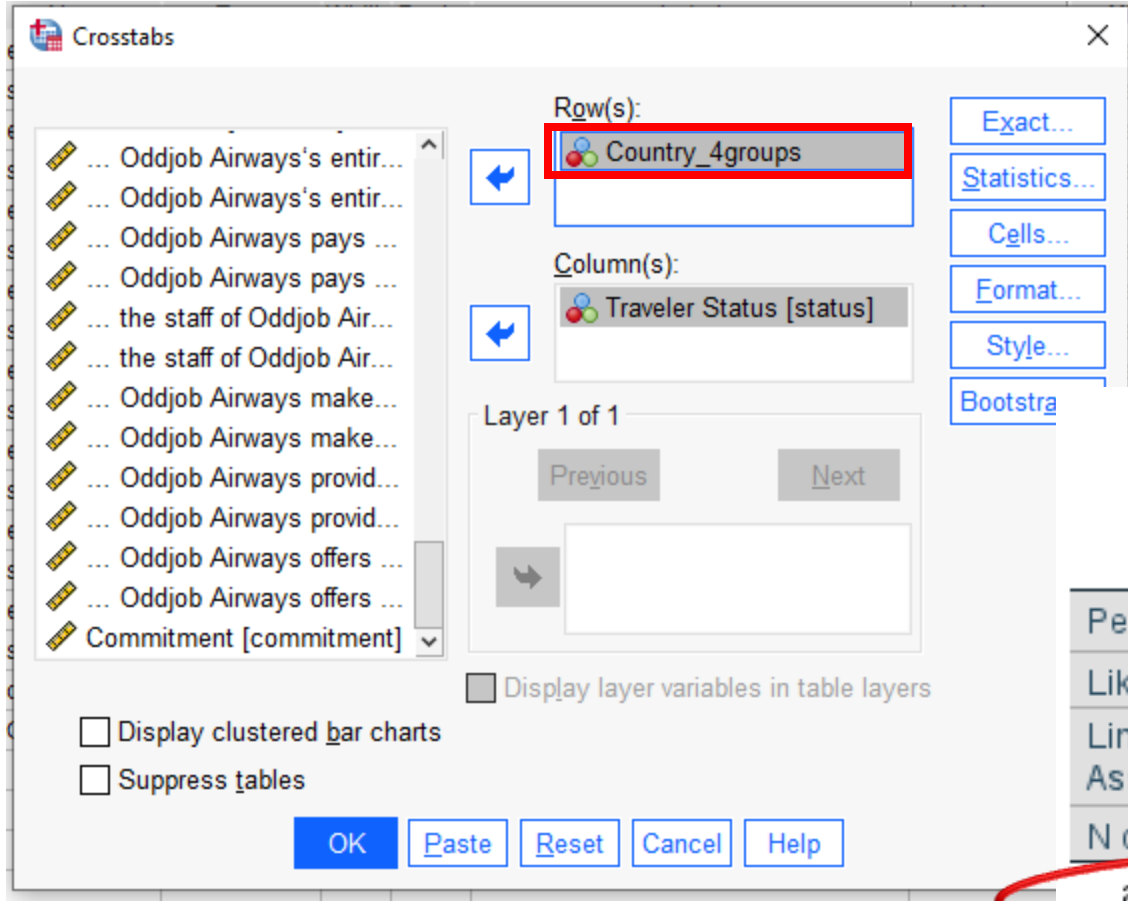
Value Labels:

Value	Label
1.00	Germany
2.00	Switzerland & France
3.00	Austria
5.00	USA

OK Reset Cancel Help

commitment	Numeric	8	2	Commitment	None
Country_4g...	Numeric	8	2		None

Chi-Square Analysis



H_0 : Country and traveler status are NOT associated
 H_1 : Country and traveler status are associated

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	35.638 ^a	6	<.001
Likelihood Ratio	37.776	6	<.001
Linear-by-Linear Association	3.905	1	.048
N of Valid Cases	1065		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 9.00.

Chi-Square Analysis

Country_4groups * Traveler Status Crosstabulation

			Traveler Status			
			Blue	Silver	Gold	Total
Country_4groups	Germany	Count	438	168	89	695
		Expected Count	441.8	159.9	93.3	695.0
		% within Country_4groups	63.0%	24.2%	12.8%	100.0%
	Switzerland & France	Count	36	20	11	67
		Expected Count	42.6	15.4	9.0	67.0
		% within Country_4groups	53.7%	29.9%	16.4%	100.0%
	Austria	Count	51	36	21	108
		Expected Count	68.7	24.8	14.5	108.0
		% within Country_4groups	47.2%	33.3%	19.4%	100.0%
	USA	Count	152	21	22	195
		Expected Count	124.0	44.9	26.2	195.0
		% within Country_4groups	77.9%	10.8%	11.3%	100.0%
Total	Count	677	245	143	1065	
	Expected Count	677.0	245.0	143.0	1065.0	
	% within Country_4groups	63.6%	23.0%	13.4%	100.0%	

Symmetric Measures

		Value	Approximate Significance
Nominal by Nominal	Contingency Coefficient	.180	<.001
N of Valid Cases		1065	

- Contingency Coefficient: Quantify the strength of association between two categorical variables. 0 (no association) to 1 (perfect association)

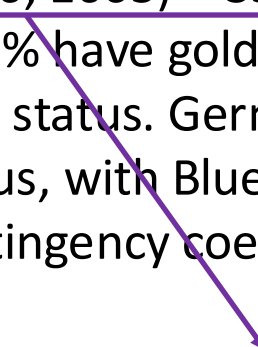


Chi-Square Analysis

How to report the results of Chi-Square tests?

Example

In this test, we hypothesized that there is an association between the home country and the traveler's status. The result of a Chi-Square test shows a significant association between these two variables ($\chi^2(6, 1065) = 35.638, p < .001$). It was found that among USA travelers, 77.9% have blue status, and 11.3% have gold status. In contrast, among Austrian travelers, 47.2% have blue status, and 19.4% have gold status. Germany and Switzerland (combined with France) show similarities in terms of traveler status, with Blue status German customers accounting for 63% of the German customers. The contingency coefficient (C) is .180.



χ^2 (degree of freedom, sample size) = chi-square value

Note: Remember to interpret your crosstab in layman's terms if there is a significant association.



Recap: Testing for Associations (Part I)

Types of Test	Purpose of Test
Chi-Square Analysis (X^2)	Test for relationship between two <u>nominal</u> variables.
Spearman Rank Order Correlation Coefficient	Test for relationship between two variables where at least one is <u>ordinal</u> .
Pearson Product Moment Correlation Coefficient	Test for relationships between two <u>interval and/or ratio</u> variables.



Types of Regression Analysis

Types of Test	Purpose of Test
Bivariate Regression Analysis	Analyze the linear relationship between <u>an</u> independent variable and <u>an interval/ratio</u> dependent variable.
Multiple Regression Analysis	Analyze the linear relationship between <u>multiple</u> independent variables and <u>an interval/ratio</u> dependent variable.



Regression Analysis

Importance of regression analysis

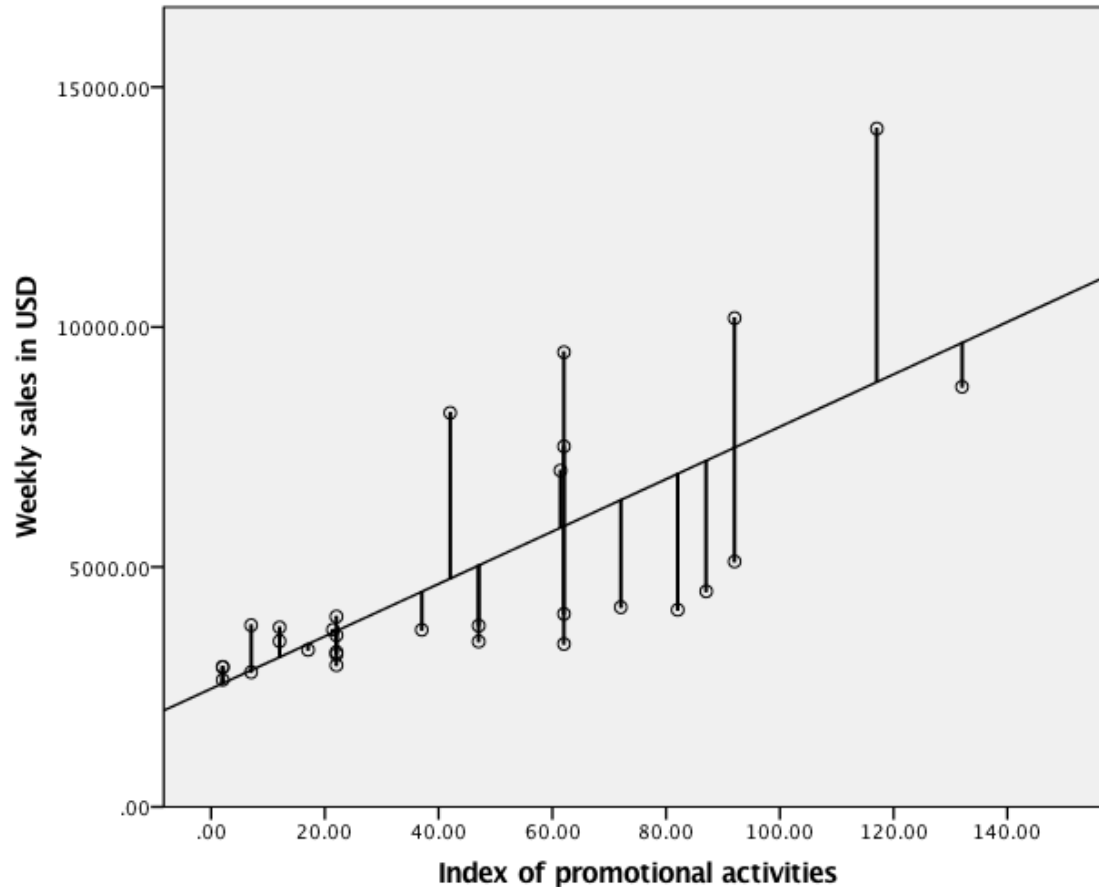
- Regression analysis is one of the most important methods in quantitative market research
- Marketing and strategy departments of major companies (e.g. Procter & Gamble, BMW Group, Nestlé) use regression analysis regularly for decision-making

Regression analysis can

1. Indicate if independent variables have a significant relationship with a dependent variable.
2. Indicate the relative strength of different independent variables' effects on a dependent variable.
3. Make predictions.

* In this lecture we introduce linear regression (OLS)

The Regression Analysis Estimate a Line Using the Observations



The regression estimation procedure attempts to fit a best possible line. This line is optimal if the squared distances to all observation points are minimized. Thereby:

1. Positive and negative deviances don't balance each other
2. Bigger deviances are weighted disproportionately high

If the assumptions of the regression are harmed, the line might not be the true line!

Understanding Regression Analysis

Notation

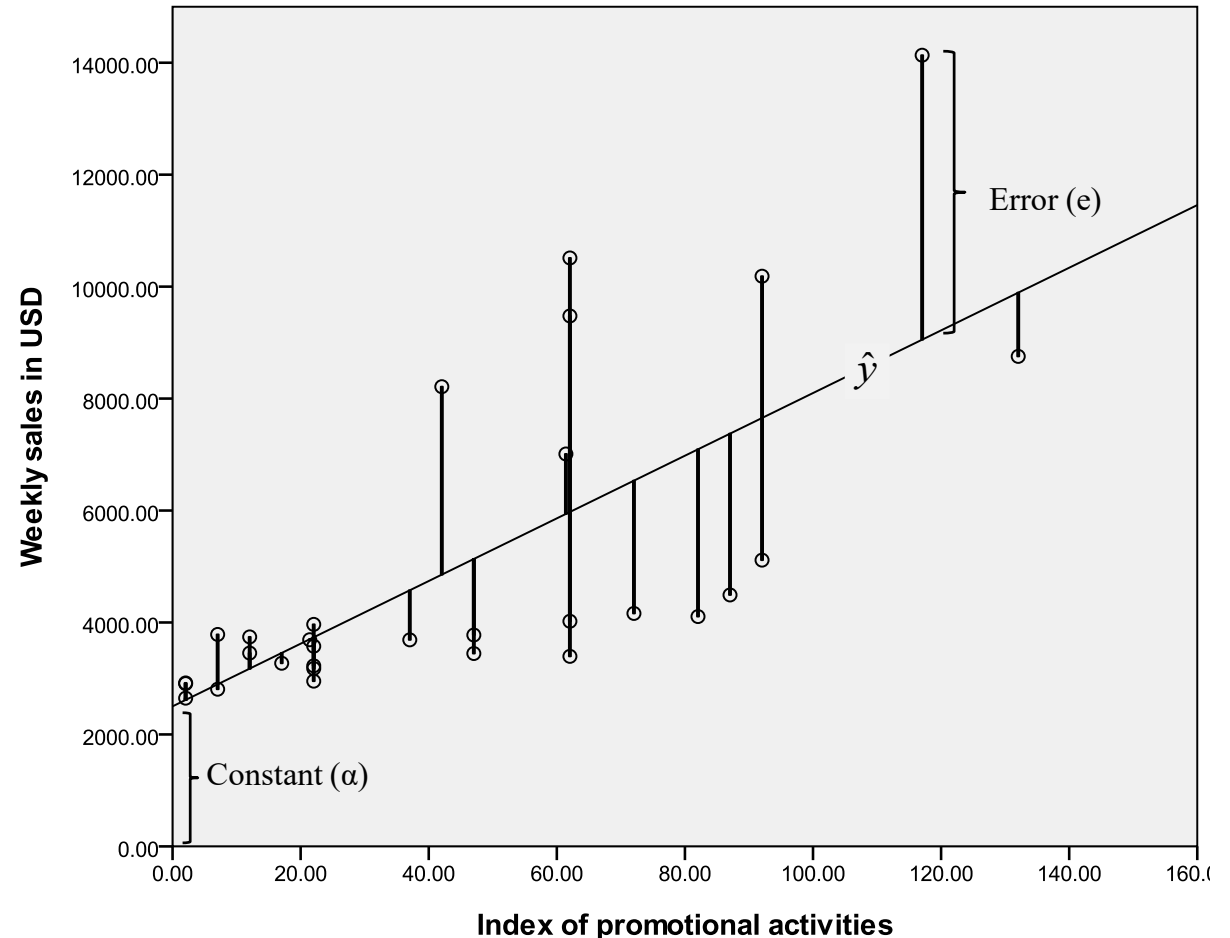
$$Y = \alpha + \beta_1 X_1 + e$$

Y	dependent variable
α	Intercept
β_1	Regression parameter
x_1	Independent variable
e	Error or residual

For multiple regression:

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + e$$

Predicted (or estimated) value



Bivariate Regression Analysis

Example

40	s8	Numeric	8	0	... Oddjob Airways offers a comfortable on-board experience.	{1, Very un... }
71	commitment	Numeric	8	2	Commitment	None

Value ▾	Label
1	Very unsatisfied
100	Very satisfied

Value ▾	Label
1	Fully disagree
7	Fully agree

- Two interval scaled variables → Bivariate Regression
 - Significance level: 5%
 - Independent variable: satisfaction with on-board experience (s8)
 - Dependent variable: commitment (commitment)
 - Hypothesis: there is a linear association between satisfaction with on-board experience and commitment.
- (*commitment is the average of 3 commitment variables (com1, com2, com3) measured on a 7-point Likert scale)



Bivariate Regression Analysis

■ $y = b_0 + b_1 x + e$

$H_0: b_1 = 0$

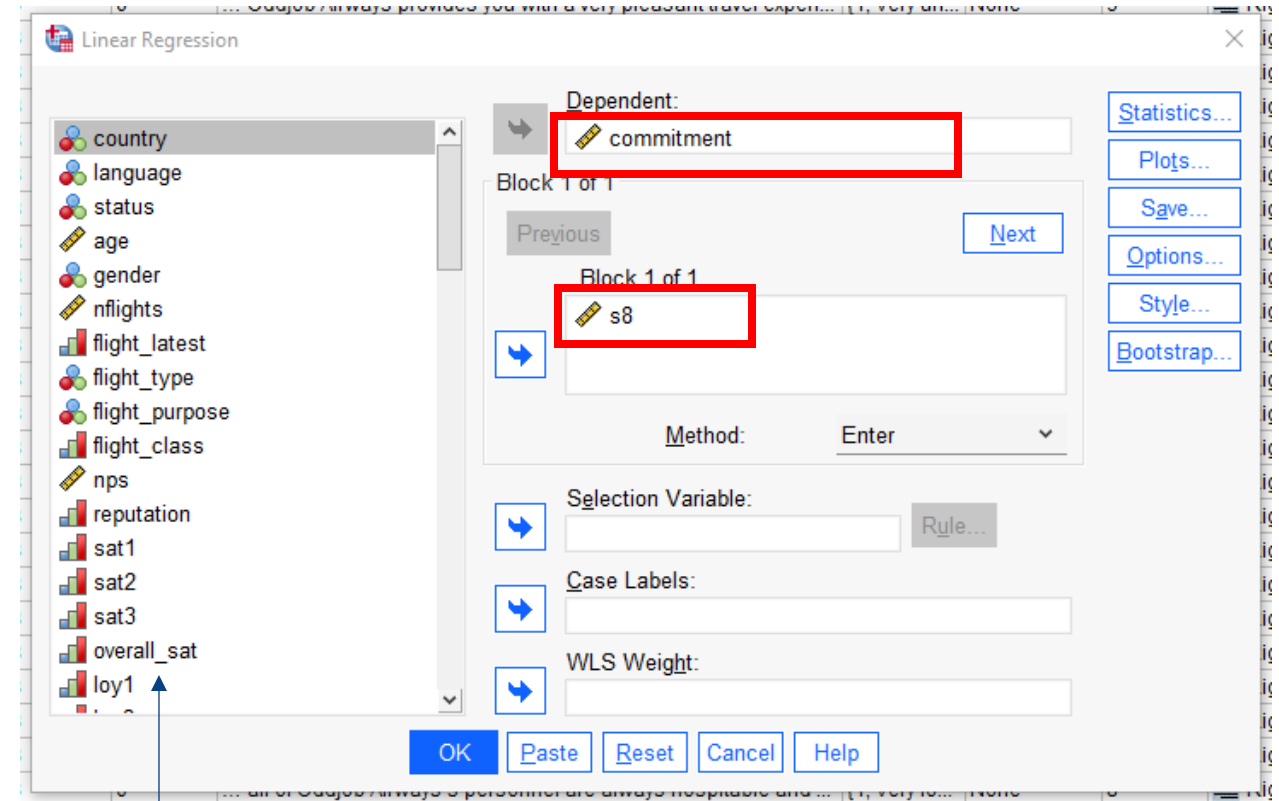
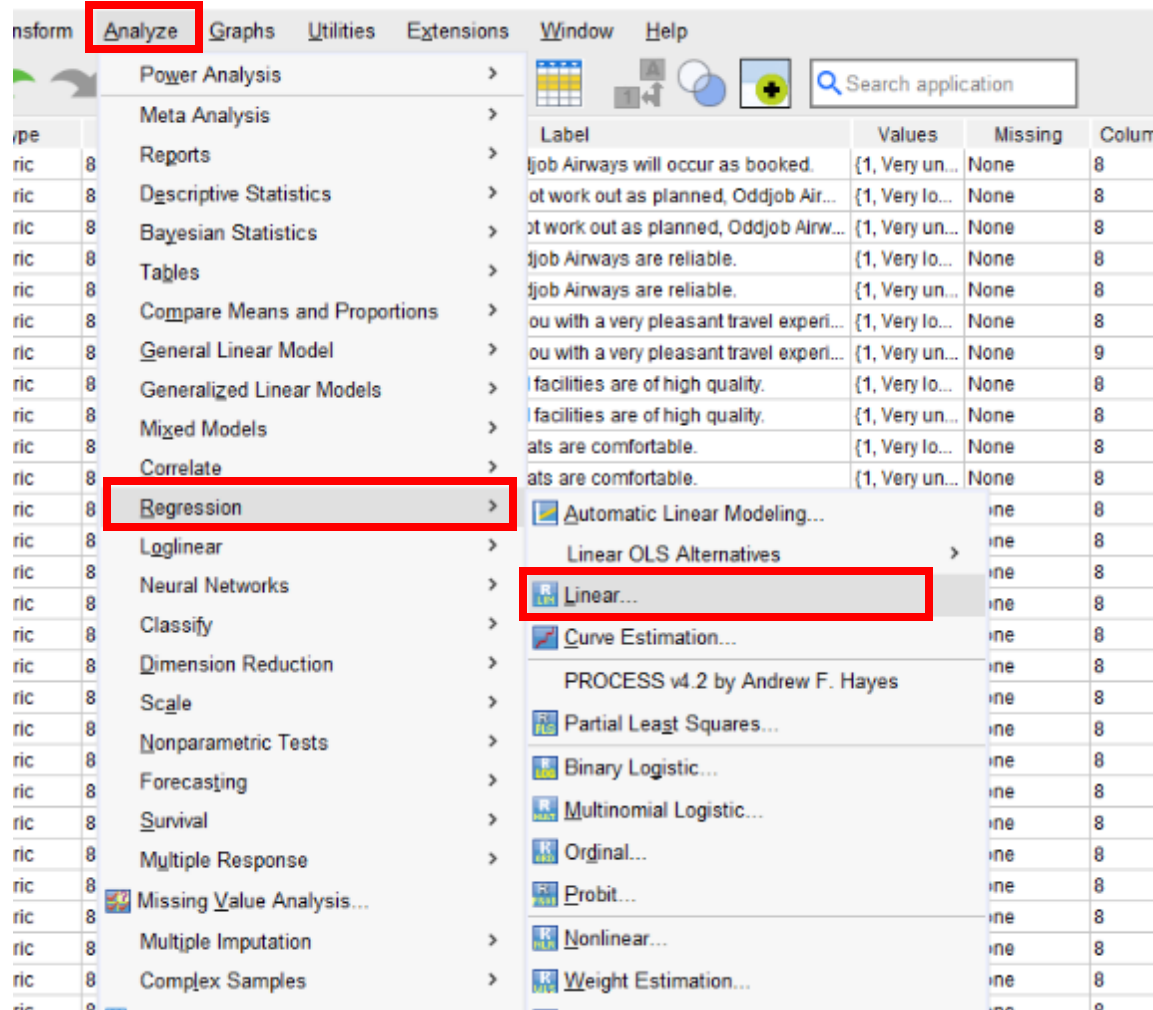
$H_1: b_1 \neq 0$

Respondents	x	y
1	2	2
2	2	2
3	2	2
4	1	3
5	2	3
6	5	7
7	6	5
8	5	6
9	4	4
10	4	5



Bivariate Regression Analysis

Analyze → Regression → Linear → Select dependent variable (to dependent) and independent variable (to block)



*Tip
Right click any variable and
select 'Display variable names'

Bivariate Regression Analysis

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	2.035	.138		14.778	<.001
	... Oddjob Airways offers a comfortable on-board experience.	.037	.002	.460	16.636	<.001

a. Dependent Variable: Commitment

Unstandardized Coefficient “B” in the Coefficient Table: The strength of the association between x and y.

$$H_0: b_1 = 0$$

$$H_1: b_1 \neq 0$$

$$y = b_0 + b_1 x + e$$

$$y = 2.035 + 0.037x + 1.546$$

Bivariate Regression Analysis

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.460 ^a	.211	.211	1.54566

Error term

a. Predictors: (Constant), ... Oddjob Airways offers a comfortable on-board experience.

R Square:

- Measures how well the regression line explains the real data points.
- Shows the percentage of variance in the dependent variable explained by the independent variable.
- A model is considered to have a good fit if R Square is .20 or above.
- R Square helps you assess whether you've constructed a good hypothesis.



Bivariate Regression Analysis

How to report the results of Bivariate Regression Analyses?

Example

In this test we hypothesized that satisfaction with on-board experience affects commitment. The result of a Bivariate Regression Analysis shows that satisfaction with on-board experience has a significant impact on commitment ($B = 0.037$, $p < .001$). Thus the hypothesis is supported.